■ Mushroom Business Management & E-Commerce Platform (Mushroom ERP)

Document 05: REST API Specification & Endpoint Contracts

Project Name: SporeERP
Document Code: DOC-05-API-SPEC
Base URL: `https://app.sporemushroom.com/api/v1` (Production) / `http://localhost:3000/api/v1` (Local)
Authentication: Bearer JWT / HTTP-Only Session Cookie (`__Secure-authjs.session-token`)
Data Format: JSON (`application/json; charset=utf-8`)

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1. Global API Standards & Error Envelope

All responses adhere to a consistent JSON envelope structure:

1.1 Success Response Envelope

1.2 Error Response Envelope (RFC 7807 Compliant)

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2. API Endpoint Directory by Domain

2.1 Authentication & Session (`/api/auth`)

Method	Endpoint	Access	Description
`POST`	`/api/auth/register`	Public	Register new customer profile.
`POST`	`/api/auth/callback/credent...`	Public	Authenticate user via email + password.
`GET`	`/api/auth/session`	Authenticated	Retrieve active session claims and permissions.
`POST`	`/api/auth/forgot-password`	Public	Trigger password reset verification email.

2.2 Product Catalog (`/api/v1/products`)

`GET /api/v1/products`

Fetches public product catalog with category, price range, and availability filters.
* **Query Parameters:**
* `category` (string, optional) — Category slug.
* `status` (string, default: `PUBLISHED`) — `DRAFT`, `PUBLISHED`, `OUT_OF_STOCK`.
* `page` (int, default: 1), `limit` (int, default: 20).
* **Response `200 OK`:**

`POST /api/v1/products`

Creates a new catalog product (Requires `catalog:product:create` permission).
* **Request Body (Zod Validated):**

- **Response `201 Created`:** Returns created product record with assigned UUID.

2.3 Customer Orders & Checkout (`/api/v1/orders`)

`POST /api/v1/orders`

Executes atomic order creation, reserves stock, and prepares payment initiation.
* **Request Body:**

- **Transactional Behavior:**
  1. Checks `available_stock >= requested_quantity` under row-level lock (`FOR UPDATE`).
  2. Creates `Order` with `order_status = 'PENDING'` and `payment_status = 'UNPAID'` .
  3. Increments `reserved_stock` and decrements `available_stock` .
  4. Generates Razorpay Order ID and returns checkout credentials.
- **Response `201 Created` :**

`PATCH /api/v1/orders/:id/status`

Transitions order fulfillment status (Requires `orders:manage` permission).
* **Request Body:**

- **Response `200 OK` :** Updates `order_status` , appends entry to `order_status_history` , and dispatches notification.

2.4 Payments & Webhook (`/api/v1/payments`)

`POST /api/v1/payments/webhook`

Webhook endpoint for payment gateway callbacks (Razorpay).
* **Headers:** `x-razorpay-signature: hex_hmac_sha256_hash`
* **Handling Logic:**
1. Verifies HMAC-SHA256 signature using secret `RAZORPAY_WEBHOOK_SECRET` .
2. On `payment.captured` : Updates `payment_status = 'PAID'` , `order_status = 'CONFIRMED'` .
3. Logs `InventoryTransaction` of type `CUSTOMER_SALE` .
4. Triggers order confirmation email to customer.
* **Response `200 OK` :** `{"status": "ok"}`

2.5 Mushroom Cultivation Batches (`/api/v1/production`)

`POST /api/v1/production/batches`

Initiates a new cultivation batch (Requires `production:batch:create`).
* **Request Body:**

- **Response `201 Created`:** Creates batch record, automatically generates 80 bed records (`BED-001` through `BED-080`), and deducts 25kg spawn and 100kg straw from Raw Materials inventory.

`PATCH /api/v1/production/batches/:id/status`

Updates batch biological status (`SPAWNING` -> `INCUBATION` -> `FRUITING` -> `COMPLETED`).

2.6 Daily Harvest Logging (`/api/v1/harvest`)

`POST /api/v1/harvest`

Logs fresh picked harvest and syncs finished goods inventory in real time.
* **Request Body:**

- **Atomic Execution:**
  1. Creates `Harvest` record with grade distribution.
  2. Updates `ProductionBatch.total_harvested_kg += 28.50`.
  3. Increments finished goods `InventoryItem` for Fresh Oyster Mushrooms by `26.50 kg` (Grade A + Grade B).
  4. Increments raw material `InventoryItem` for Dehydration by `1.50 kg` (Grade C).
  5. Inserts `InventoryTransaction` of type `HARVEST_INJECTION`.
- **Response `201 Created`:**

2.7 Expenses & Financial Reporting (`/api/v1/expenses` & `/api/v1/reports/pnl`)

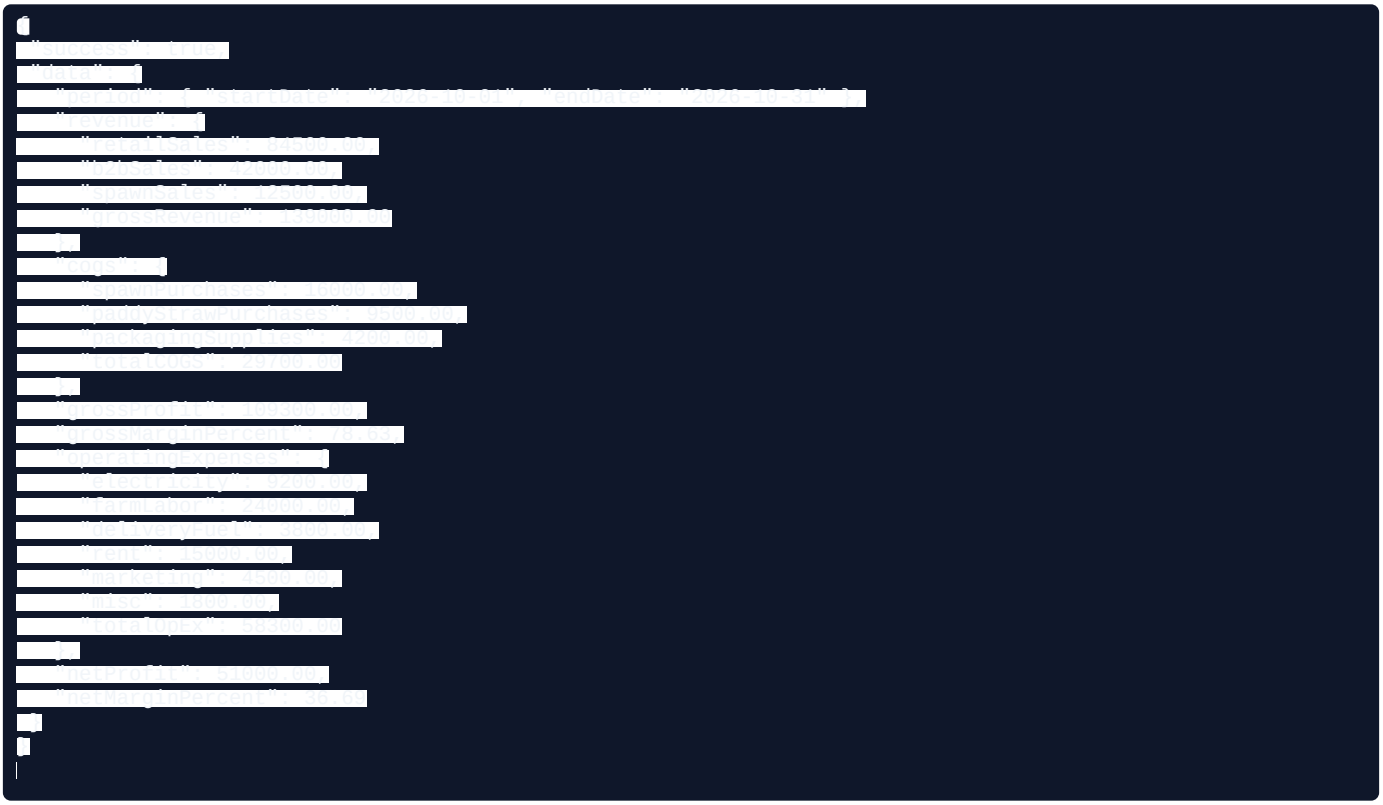
`POST /api/v1/expenses`

Records business operating cost or raw material payment.
* **Request Body:**

- **Response `201 Created`:** Returns expense record.

`GET /api/v1/reports/pnl`

Real-time Profit & Loss statement calculation.

* **Query Parameters:** `startDate=2026-10-01&endDate=2026-10-31`

* **Response** `200 OK`**:**